OXFORD
INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL GCSE
Biology

9201/2 – Paper 2

Mark scheme

9201

June 2018

Version/Stage: 1.0 Final

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

2.1 In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.

2.2 A bold **and** is used to indicate that both parts of the answer are required to award the mark.

2.3 Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

2.4 Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of error / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	**Response**	**Marks awarded**
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	**Response**	**Marks awarded**
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(…..) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do <u>not</u> accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

4. Level of response marking instructions

Extended response questions are marked on level of response mark schemes.

- Level of response mark schemes are broken down into levels, each of which has a descriptor.
- The descriptor for the level shows the average performance for the level.
- There are two marks in each level.

Before you apply the mark scheme to a student's answer, read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

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Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO/Spec. Ref.
01.1	A = nucleus B = chromosome		1 1	AO1 3.3.1a 3.5.2a
01.2	gametes		1	AO1 3.5.2 f
01.3	fertilisation		1	AO1 3.5.2 i
01.4	mitosis		1	AO1 3.5.2 d
01.5	10 (years)	allow 10 to 10.7 years allow 10 years 0 months to 10 years 9 months	1	AO3 3.5.2 d
01.6	15 (years)	allow 14.6 to 15 years allow 14 years 9 months to 15 years 0 months	1	AO3 3.5.2 d
01.7	replace (worn out/damaged) cells	allow repair	1	AO3 3.5.2 d
Total			**8**	

Question	Answers	Extra information	Mark	AO/Spec. Ref.
02.1		an answer of 5.2 scores 2 marks		AO2 6.3 (11,4)
	6 – 0.8		1	
	5.2	5,200,000,000 scores one mark	1	
02.2	bar drawn correct height	Allow +/- half small square	1	AO2 6.3 (9)
02.3	any **two** from: • medicines to treat/cure disease • lots/more than enough food • heating to keep warm	allow other sensible suggestions eg improvements in pre/postnatal care **or** improved waste disposal **or** water treatment if no other marks gained allow increase in standard of living or better hygiene for **1** mark.	2 max	AO1 AO3 3.3.4a
02.4	Carbon dioxide		1	AO1 3.3.4d
02.5	increase in temperature (of the surface) of the Earth		1	AO1 3.3.4 d

02.6		less needed at least once to score full marks ignore CO_2 produced by trucks, chain saws or burning		AO1 AO2 3.3.3e 3.3.4d
	(fewer trees means) less/no photosynthesis		1	
	less/no carbon dioxide absorbed (from the environment) **or** more carbon dioxide in atmosphere **OR** more carbon dioxide in atmosphere (1) prevents infra-red waves/energy escaping into space (1) **or** energy reflected/re-radiated back to earth (warming it up)	allow removed / taken in	1	

02.7	dead plant matter decayed/broken down/digested/decomposed		1	AO1 AO2 3.3.3e 3.3.4d
	by microorganisms	allow bacteria/fungi/decomposers ignore detritivores	1	
	which respire		1	
	releasing (more) carbon dioxide	allow methane	1	

Total			**13**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
03.1	any **two** from: • ribosome(s) • (cell) wall • (cell) membrane • cytoplasm		2 max	AO1 3.1.1 b, c
03.2	prevents growth of **OR** kills bacteria		1	AO1 3.4.7 f
03.3	(random) mutations occur (giving resistance) the <u>resistant bacteria</u> (survives and) reproduces/replicates population (of resistant strain) increases due to lack of competition or due to non-resistant killed by antibiotic		1 1 1	AO1 AO2 3.4.7h,i
03.4	to produce a plate with only one species	allow to prevent contamination from other bacteria or other species may affect results.	1	AO4 6.2
03.5	optimum/best temperature to allow (these) bacteria to grow/replicate		1	AO4 6.2

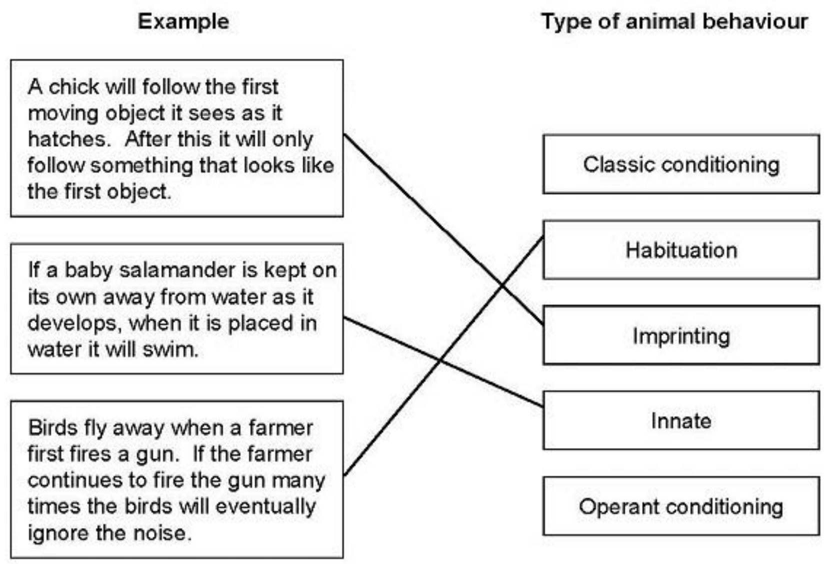
03.6	any **two** from: • wash hands with (disinfectant) soap • clean surfaces with disinfectant • wear safety goggles • keep Petri dish sealed • dispose of waste in biohazard bag **or** by using pressure cooker/autoclave/heating to 120°C • incubate at 20−25°C	allow alcohol allow wear disposable gloves allow alcohol allow eye protection allow do not eat **or** drink	2 max	AO4 6.2

03.7	diameter 15 **and** 16 (any order) mean diameter = 15.5	allow correct calculation for mean based upon student's own measurements.	1 1	AO2 6.2 6.3

03.8	<u>In support:</u> both solutions (blueberry juice and coconut oil) produced a clear area **or** prevented growth of bacteria with at least medium antibacterial activity. <u>Qualified:</u> only coconut oil showed high antibacterial activity or blueberry juice (only) showed a medium level of activity. <u>Use of data</u>: (because) blueberry juice has a (medium antibacterial activity with a) diameter of 11.5 mm which is between 9 and 13 mm **or** (because) coconut oil has a (high antibacterial activity with a) diameter of 15.5 mm which is greater than 13 mm **or** antibiotic produced the greatest clear area by at least 2.5mm **or** 16% **or** 6.5mm **or** 57% more than blueberry juice.	allow ecf from student's answer in 03.7 allow any correct comparison using data	1 1 1	AO3 6.2 6.3 (3)

03.9	any **two** from: • first scientist did not repeat the test **or** anomalies unknown with only one Petri dish (for first scientist) • (clear areas not symmetrical so) depends on where the diameter was measured • concentration/volume of solutions may differ • may have used a different type/strain of bacteria • different sources/brands of plant extract • paper discs may have been different sizes	If not other marks gained allow used different equipment for **one** mark.	2 max	AO2 6.2 6.3

Total			**17**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
04.1		**Extra information** one mark per correct response more than one line from an example will cancel the mark	3	AO2 3.4.6c

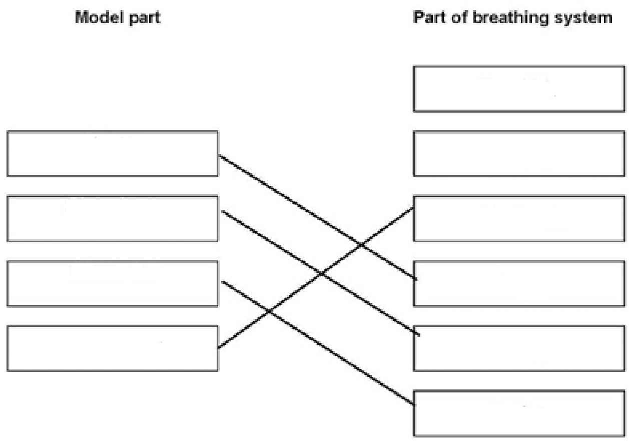
04.2	any **four** from: • use sample size of 20 or more rats • (randomly) separate rats into 2 groups • place 1 or 4 pieces of food at the finish according to the group • put one rat at a time at the start (of T- maze) • record if the rat reaches the food or not • repeat every day for 5 days • a relevant control variable • calculate percentage for each group (which took the correct route)	example of a control variable: same type/same size of food pieces **or** same diet prior to test/starved **or** same environmental noise **or** same age/type of rats	4 max	AO2 6.1

04.3	rat learns to carry out activity/action associated with reward/food by trial and error		1 1 1	AO1 AO2 3.4.6c

Level 3: Relevant points (reasons/causes) are identified, given in detail and logically linked to form a clear account.	**5-6**	
Level 2: Relevant points (reasons/causes) are identified, and there are attempts at logically linking. The resulting account is not fully clear.	**3-4**	
Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	**1-2**	
No relevant content	**0**	

04.4	**Indicative content** stickleback example – mouth brooding **or** young kept in mouth for first week Advantages: • protection in early stages **or** first week increases survival rate (compared to turtle) • resources required from parent **or** risk to parent are reduced Disadvantages: • only protected for first week • low survival rate (compared to mammal) Turtle example – egg laying **or** egg contains food for developing offspring Advantages: • no risk to parent **or** no further time **or** resources spent on offspring Disadvantages: • very low survival rate **or** lowest survival rate • most die **or** get eaten • parent has no way of increasing likelihood of offspring surviving • relies upon innate behaviour **or** no learning from parent Lion example – early development within mother (uterus) Advantages: • high level of parental care increases chances of offspring surviving **or** passing on genes to next generation **or** highest survival rate • offspring have a reliable source of food **or** mother produces milk • protection from predators • taught how to avoid predators and where to find food etc. Disadvantages • mother at risk from starvation or disease **or** predators • uses up a lot of mother's time and resources.		AO2 AO3 3.4.6b

Total			**16**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
05.1			4	AO2 3.2.5a
05.2	increase in volume (of chest/dome) pressure reduced/pressure less inside compared to outside.	allow volume of lungs ignore pressure changes unspecified	1 1	AO2 3.25b
05.3	$\sqrt{\frac{179 \times 70}{3600}}$ **or** 1.865.. 1.87 4675	an answer of 4675 scores 3 marks an answer of 1.87 scores 2 marks allow correct calculation from incorrect BSA allow 4664 (.0617…) from using unrounded BSA in calculation	1 1 1	AO2 6.3 (4,10)
05.4	the larger the vital capacity the smaller the increase in breathing rate	allow converse allow negative correlation	1	AO3 6.1 6.3 (12)

	points are scattered **or** not a very clear relationship	allow large variation in data	1	
	appropriate use of data from graph eg increase for largest capacity/5500 was 6 whereas smallest capacity/3700 was 12	allow other correct use of data	1	

05.5	(larger lung capacity) more oxygen enters per breath **or** same volume of oxygen enters in fewer breaths	allow converse reasoning for all points about carbon dioxide	1	AO2 AO3 3.2.6a,g,h 6.1
	(therefore) breathing rate does not have to increase as much to allow for increased oxygen requirement		1	

05.6	any **three** from: • spirometer removes any errors from calculation **or** • the result is read directly from the display • spirometer measures volume rather than a calculation based on height and mass • spirometer takes into account general health • spirometer takes into account physical characteristics eg body types/age/gender • spirometer takes into account fitness levels • breathing rate for 4000 fits the general trend (whereas 5000 does not) implying spirometer is more accurate	allow ref to eg named respiratory disease	3 max	AO4 6.1

Total			**17**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
06.1	$\frac{20}{27} \times 100$ **or** $\frac{47 - 27}{27} \times 100$ **or** 74.07 …	an answer of 74.1 (%) scores 2 marks	1	AO2 6.3 (12,3)
	74.1(%)		1	
06.2	(more carbon dioxide taken in) greater rate of photosynthesis or more glucose produced		1	AO1 AO2 3.2.1d 3.2.6f
	(so more glucose produced) for respiration or energy release	do not allow make/create/produce/generate energy	1	
	to produce proteins/cellulose/other organic molecules (for growth)		1	

06.3	**Light** For: (light intensity of 80 000 lux) had the greatest (maximum) % increase Against: productivity values are still increasing suggesting light intensity is a limiting factor (so farmer could use a higher light intensity) **Carbon dioxide** For: greatest increase in rate/increase in productivity is greatest between 0.04% and 0.06% Against: not the highest (productivity) value **or** 0.08/0.10% are higher **Profit** cost of providing more carbon dioxide and/or light might be greater than the gain in yield/productivity.	allow comparative use of figures eg 80 000 was over twice the rise compared to 10 000 allow rate of increase slows between 0.06% and 0.08%	1 1 1 1 1	AO3 3.2.1c 6.1

Total			**10**

Question	Answers	Extra information	Mark	AO/Spec. Ref.
07.1	any **three** from: • no end walls - enables water and/or (mineral) ions to flow through easily • no cytoplasm - gives room for water and/or (mineral) ions to pass through • no cytoplasm/cell components for chemical reactions to take place • thickened walls – supports/strengthens xylem • small hole(s) - allows water and/or (mineral) ions to pass in or out of vessel	allow reference to lignin or lignin rings providing strength/support	3 max	AO2 AO3 3.1.4a 3.2.2f
07.2	any **three** from: • needs nucleus to control cell activities • (many) mitochondria to carry out respiration • (which) releases energy/ATP • for active transport of sugars • ribosomes for protein synthesis	do not allow make/create/produce/generate energy	3 max	AO3 3.1.4a 3.2.2f

07.3	plant A shows sugars can move up and down the plant/stem plant B shows that movement of sugars requires phloem (to be intact) **or** sugars cannot move above ring where there is no phloem plant C provides evidence that the sugar required for growth/living cells is provided by the phloem (and not by the roots or xylem)	If no other mark awarded allow **1** mark for radioactive CO_2 goes into the sugar/glucose which is transported in the phloem	1 1 1	AO4 3.1.4a 3.2.2f

Total			**9**